ConstructPro

Software Requirements Specification (SRS)

By

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Table of Contents

List of Tables v

List of Figures vi

Chapter 1 Introduction 1

1.1 Background 1

1.2 System overview 1

Chapter 2 Purpose 3

2.1 Purpose of the document 3

2.2 Intended audience 3

Chapter 3 Scope 4

3.1 System overview 4

3.2 System functions 5

3.3 Scope exclusions 5

Chapter 4 Definitions, Acronyms and Abbreviations 6

Chapter 5 Document Overview 8

Chapter 6 Project Scope 10

6.1 Business Objectives 10

6.2 Project Objectives 10

6.3 System Context 11

Chapter 7 Stakeholders 12

Chapter 8 Functional Requirements 14

Chapter 9 Non-Functional Requirements 17

Chapter 10 Business Rules 19

10.1 User Access Control 19

10.2 Quotation Approval Policy 19

10.3 Payment Recording Rule 19

10.4 Project Milestone Management 19

10.5 Client Portal Access 19

10.6 Quotations per Project 19

Chapter 11 Constraints 20

11.1 Technology Constraints 20
11.2 Integration Constraints 20
11.3 Security Constraints 20
11.4 Deployment Constraints 20
Chapter 12 Assumptions 21
12.1 User Training 21
12.2 Internet Connectivity 21
12.3 Availability of Historical Data 21
12.4 Organizational Support 21
Chapter 13 Use Case Diagrams 22
13.1 User & Role Management Use Case Diagram 22
13.2 Lead & Quotation Management Use Case Diagram 22
13.3 Project Management Use Case Diagram 23
13.4 Finance Management Use Case Diagram 23
13.5 Analytics & AI Prediction Use Case Diagram 24
13.6 Document Management Use Case Diagram 24
Chapter 14 Use Case Descriptions 25
14.1 UC 01 — Manage Users 25
14.2 UC 02 — Capture Sales Lead 26
14.3 UC-03 — Generate Quotation 26
14.4 UC-04 — Convert Quotation to Project 27
14.5 UC-05 — Manage Project Milestones 28
14.6 UC-06 — Record Client Payment 28
14.7 UC-07 — View KPI Dashboard 29
14.8 UC-08 — Upload Project Documents 30
14.9 UC-09 — Schedule Meeting 30
14.10 UC-10 — AI Risk Prediction 31
Chapter 15 Activity Diagrams 32
15.1 Lead, Quotation & Project Workflow 32
15.2 Project Management Workflow 33
15.3 Payment Management Activity 34
15.4 KPI Dashboard Workflow 35
15.5 AI Risk Prediction Workflow 36
Chapter 16 External Interface Requirements 37

16.1 User Interface Requirements 37
16.2 Hardware Interface Requirements 37
16.3 Software Interface Requirements 37
16.4 Communication Interface Requirements 38
Chapter 17 Data Requirements 39
17.1 Data Entities 39
17.2 Data Storage 39
17.3 Data Integrity 39
Chapter 18 Alternative Solutions Considered 40
18.1 Manual Management Systems 40
18.2 Commercial ERP Systems 40
18.3 Custom ERP Solution 40
Chapter 19 Feasibility Study 41
19.1 Technical Feasibility 41
19.2 Economic Feasibility 41
19.3 Operational Feasibility 41
19.4 Schedule Feasibility 41
Chapter 20 Risk Analysis 42
20.1 Risk Register 42
20.2 Risk Categories 43
Chapter 21 References 45
Client Requirement Sign-off 46

List of Tables

Table 1: Definitions, Acronyms and Abbreviations 7
Table 2: Stakeholders 13
Table 3: Functional Requirements 15
Table 4: Non-Functional Requirements 18
Table 5: UC01 - Manage Users 25
Table 6: UC02 - Capture Sales Lead 26
Table 7: UC03 - Generate Quotation 27
Table 8: UC04 - Convert Quotation to Project 27
Table 9: UC05 - Manage Project Milestones 28
Table 10: UC06 - Record Client Payment 29
Table 11: UC07 - View KPI Dashboard 29
Table 12: UC08 - Upload Project Documents 30
Table 13: UC09 - Schedule Meeting 31
Table 14: UC10 - AI Risk Prediction 31
Table 15: Risk Register 43

List of Figures

Figure 1: User & Role Management Use Case Diagram 22
Figure 2: Lead & Quotation Management Use Case Diagram 22
Figure 3: Project Management Use Case Diagram 23
Figure 4: Finance Management Use Case Diagram 23
Figure 5: Analytics & AI Prediction Use Case Diagram 24
Figure 6: Document Management Use Case Diagram 24
Figure 7: Lead, Quotation & Project Workflow 32
Figure 8: Project Management Workflow 33
Figure 9: Payment Management Activity 34
Figure 10: KPI Dashboard Workflow 35
Figure 11: AI Risk Prediction Workflow 36

Chapter 1 Introduction

1.1 Background

Construction companies often manage project information, client leads, quotations, and financial records using manual processes or disconnected software tools. As organizations grow, these approaches lead to inefficiencies such as delayed communication, inconsistent data, and limited visibility into project progress and financial performance. For companies like Ishara Homes Pvt. Ltd., managing the entire lead-to-project lifecycle efficiently requires a centralized digital system that integrates sales, project management, and financial tracking.

Enterprise Resource Planning (ERP) systems address these challenges by integrating organizational processes into a unified platform, allowing departments to share information in real time and improve decision-making capabilities [1]. Implementing such a system can significantly improve operational efficiency and reduce risks associated with manual data management.

1.2 System overview

The system proposed in this project is ConstructPro ERP, a cloud-based enterprise resource planning platform designed to manage the complete workflow of construction projects. The system integrates several functional modules, including lead management, quotation generation, project tracking, invoice management, document storage, and analytics reporting.

The architecture of ConstructPro ERP follows a modern multi-tier design consisting of a Next.js frontend application, a Nest.js backend API server, and a PostgreSQL data layer for persistent storage. The system also includes several supporting services such as document generation, calendar synchronization, and file storage. External services such as the Google Calendar API, Google OAuth and Cloudinary are integrated to support scheduling, sign-in and document storage.

In addition to traditional ERP features, the system incorporates an AI forecasting service based on Retrieval-Augmented Generation (RAG) to analyze historical project data and generate predictive insights related to revenue trends and project risks. This capability enhances the system by transforming it from a basic management tool into a data-driven decision support platform.

Overall, ConstructPro ERP aims to provide a scalable, secure, and intelligent solution that improves collaboration between departments and enables efficient management of construction projects.

Chapter 2 Purpose

2.1 Purpose of the document

This Software Requirements Specification (SRS) document serves the following purposes:

- To provide a complete, clear, and formally structured description of the functional and non-functional requirements of the ConstructPro ERP system, which is designed to manage the lead-to-project lifecycle of construction projects at Ishara Homes Pvt. Ltd.
- To serve as the official agreement between the client (Ishara Homes Pvt. Ltd.), the development team, and the academic supervisor regarding the scope, features, and operational behavior of the system.
- To establish the baseline requirements that will guide the subsequent phases of the project, including system design, implementation, testing, and deployment.
- To document all stakeholder-approved requirements with unique identifiers, priority levels, and implementation status, enabling requirement traceability throughout the software engineering lifecycle.

2.2 Intended audience

This document is intended for the following stakeholders:

- System developers who will implement the system
- Project managers responsible for system planning and development
- Ishara Homes management and staff who will use the system
- Academic supervisors evaluating the system design project

Chapter 3 Scope

3.1 System overview

The system described in this document is named ConstructPro ERP. It is a cloud-based enterprise resource planning system developed to support the operational processes of Ishara Homes Pvt. Ltd. The system centralizes key business activities such as lead management, quotation generation, project tracking, and financial monitoring into a single integrated platform.

ConstructPro ERP consists of several functional modules designed to support different organizational roles and operational activities.

- Sales Management Module (SMM) — enables the sales team to capture potential client leads, maintain customer information, and generate structured project quotations for construction services.
- Project Management Module (PMM) — allows project managers to convert approved quotations into active projects, define project milestones, and monitor the progress of construction activities.
- Financial Management Module (FMM) — provides functionality for recording client payments, monitoring outstanding balances, and maintaining financial records related to project transactions.
- Document Management Module (DMM) — allows secure storage and management of project-related documents such as contracts, architectural drawings, approvals, and reports.
- Analytics and Reporting Module (ARM) — provides management with real-time dashboards and key performance indicators (KPIs) such as revenue tracking, project completion status, and payment monitoring.
- AI Predictive Analysis Module (AIPAM) ― utilizes a Retrieval-Augmented Generation (RAG) approach, retrieving ERP context stored with pgvector and calling a large language model through OpenRouter, to analyze ERP data and generate insights related to potential payment delays, project risks, and operational trends.
- Client Portal (CP) — gives customers of Ishara Homes read-only access to their own projects, quotations, invoices, payments and shared documents.

3.2 System functions

Within its defined scope, ConstructPro ERP will provide the following capabilities:

- Capturing and managing client leads for potential construction projects.
- Generating project quotations, including several quotations for the same project, and maintaining quotation records.
- Converting approved quotations into active projects.
- Tracking project milestones and monitoring project progress.
- Managing project tasks and recording project expenses.
- Recording and managing client payments related to projects.
- Providing centralized document storage for project-related files.
- Generating management dashboards and analytics reports.
- Integrating scheduling functionality using Google Calendar.
- Providing predictive insights through an AI-based risk analysis system.
- Enforcing role-based access control using secure authentication mechanisms.
- Giving customers a read-only client portal for their own projects.
- Allowing staff to sign in with a Google account as well as with email and password.

3.3 Scope exclusions

The following features are not included within the scope of ConstructPro ERP:

- Integration with external accounting or payroll management systems.
- Automated procurement or inventory management of construction materials.
- Direct integration with banking systems for payment processing.
- Mobile-native application development (iOS or Android). The system will be accessible through web browsers.
- Management of physical construction equipment or on-site hardware devices.
- Contractor management, material tracking and automated notifications, which were postponed during development in favour of the lead-to-cash workflow.

Chapter 4 Definitions, Acronyms and Abbreviations

This section defines the technical terms, acronyms, and abbreviations used throughout this document to ensure a clear and consistent understanding of the ConstructPro ERP system.

Term / Acronym	Definition
ERP	Enterprise Resource Planning – a software system used to integrate and manage core business processes within an organization.
ConstructPro ERP	The cloud-based enterprise resource planning system developed for Ishara Homes Pvt. Ltd. to manage leads, projects, and financial activities.
RAG	Retrieval-Augmented Generation – an AI technique that combines information retrieval with language models to generate context-aware responses.
AI	Artificial Intelligence – technology used to enable machines to simulate human intelligence and perform tasks such as prediction and analysis.
JWT	JSON Web Token – a secure authentication method used for transmitting information between parties as a digitally signed token.
OAuth 2.0	Open Authorization – the standard used for Google sign-in and for connecting each user's Google Calendar.
Client portal	The read-only part of the system where customers view their own projects, quotations, invoices and documents.
RBAC	Role-Based Access Control – a security mechanism that restricts system access based on user roles and responsibilities.

KPI	Key Performance Indicator – measurable metrics used to evaluate business performance such as revenue and project completion rates.
API	Application Programming Interface – a set of protocols that allow different software systems to communicate with each other.
CRUD	Create, Read, Update, Delete – the four basic operations used in database management systems.
Frontend	The user-facing interface of the system through which users interact with the application.
Backend	The server-side component responsible for processing logic, managing APIs, and handling data operations.
Database	A structured collection of data used by the system to store and manage information such as users, projects, and payments.
PostgreSQL	A relational SQL database used to store structured ERP data
Next.js	A React-based web framework used to build the frontend of the system.
Nest.js	A backend web framework used with Node.js to develop the system's API services.

Table 1: Definitions, Acronyms and Abbreviations

Chapter 5 Document Overview

This Software Requirements Specification (SRS) document describes the requirements and specifications of the ConstructPro ERP system developed for Ishara Homes Pvt. Ltd. The document provides a structured description of the system's objectives, functional and non-functional requirements, system constraints, and operational expectations. It serves as a reference for stakeholders including the development team, project supervisors, and organizational management to understand the system scope and requirements before the design and implementation phases.

This Software Requirements Specification (SRS) is organized into the following chapters:

- Chapter 1 — Introduction: Provides an overview of the ConstructPro ERP system, including background information about the project, the purpose of the document, the system scope, and key definitions used throughout the document.
- Chapter 2 — Purpose: Describes the purpose of the SRS document and identifies its intended audience, including developers, stakeholders, and project supervisors.
- Chapter 3 — Project Scope: Describes the overall scope of the ConstructPro ERP system, including its functional modules, system capabilities, and the features that are included and excluded from the system.
- Chapter 4 — Definitions, Acronyms, and Abbreviations: Explains important technical terms, acronyms, and domain-specific terminology used throughout the document.
- Chapter 6 — Project Scope: Describes the objectives and operational goals of the system, including how ConstructPro ERP supports project management, financial monitoring, and organizational decision-making.
- Chapter 7 — Stakeholders: Identifies the primary stakeholders involved in the system such as management, sales staff, project managers, and finance officers, and explains their roles and interests in the system.
- Chapter 8 — Functional Requirements: Lists all functional requirements of the system using unique identifiers, including features such as lead management, quotation generation, project tracking, payment management, document storage, and analytics reporting.

- Chapter 9 — Non-Functional Requirements: Describes the quality attributes of the system, including performance, reliability, usability, security, scalability, maintainability, and portability.
- Chapter 10 — Business Rules: Defines the business policies and operational rules that govern how the ConstructPro ERP system operates within the organization.
- Chapter 11 — Constraints: Describes the technical, organizational, and implementation constraints that influence the design and development of the system.
- Chapter 12 — Assumptions: Lists the assumptions made during the requirement gathering process that may affect system design or implementation.
- Chapter 13 — Use Case Diagrams: Presents UML use case diagrams illustrating the interactions between system users and the ConstructPro ERP system.
- Chapter 14 — Use Case Descriptions: Provides detailed descriptions of the main use cases including actors, preconditions, main flows, and alternative flows.
- Chapter 15 — Activity Diagrams: Illustrates the workflow of key business processes such as lead management, quotation processing, project management, and payment tracking.
- Chapter 16 — External Interface Requirements: Describes the user interface, hardware interface, software interface, and communication interfaces used by the system.
- Chapter 17 — Data Requirements: Defines the data entities and database requirements used within the ConstructPro ERP system.
- Chapter 18 — Alternative Solutions Considered: Discusses alternative technological approaches evaluated during system analysis.
- Chapter 19 — Feasibility Study: Analyzes the feasibility of the project from technical, economic, operational, and schedule perspectives.
- Chapter 20 — Risk Analysis: Identifies potential project risks and provides mitigation strategies through a structured risk register.
- Chapter 21 — References: Lists all external documents, standards, and sources referenced in the preparation of this SRS document.

Chapter 6 Project Scope

6.1 Business Objectives

Ishara Homes Pvt. Ltd. currently manages its construction project operations using manual processes and separate spreadsheet files. Activities such as managing client leads, preparing quotations, tracking project milestones, and recording financial transactions are often handled independently across different departments. As the organization grows, these disconnected methods create several operational challenges, including delays in communication, difficulty in tracking project progress, inconsistent financial records, and limited visibility into overall business performance.

The lack of a centralized system makes it difficult for management to obtain accurate and real-time information required for decision-making. Sales teams may not have access to updated project status, finance teams may experience delays in tracking client payments, and project managers may face challenges in monitoring milestones and deadlines. Therefore, a unified digital platform is required to streamline these processes and improve operational efficiency.

6.2 Project Objectives

The primary objective of the ConstructPro ERP project is to develop a centralized system that integrates key business processes within Ishara Homes Pvt. Ltd. The system aims to improve coordination between departments, enhance data accuracy, and provide better visibility into project performance.

The system is designed to achieve the following objectives:

- Digitize the lead-to-project lifecycle by managing leads, quotations, and projects within a single platform.
- Improve project monitoring by enabling real-time tracking of project milestones and progress.
- Provide financial transparency through systematic recording of invoices and client payments.
- Enable centralized document management for project-related files such as contracts and drawings.

- Provide management with analytics dashboards and key performance indicators for monitoring business performance.
- Support intelligent decision-making through an AI-based predictive analysis module.

6.3 System Context

ConstructPro ERP is implemented as a web-based enterprise system that integrates multiple components within a modern cloud architecture. The system consists of a Next.js frontend application, a Nest.js backend API server, and a PostgreSQL database hosted on Neon for storing system data. Users interact with the system through a web interface that provides role-based access to system modules depending on their responsibilities.

The system interacts with several external services, including the Google Calendar API for scheduling project-related meetings, Google OAuth for staff sign-in, and Cloudinary for storing project documents and quotation PDFs. In addition, the platform includes an AI-based forecasting module that analyzes ERP data to generate insights related to project risks and financial trends.

Within the organizational environment, ConstructPro ERP supports multiple stakeholders, including administrators, sales teams, project managers, and finance officers, enabling them to collaborate efficiently through a centralized information system.

Chapter 7 Stakeholders

Stakeholders are individuals or groups who have an interest in the development, implementation, and operation of the ConstructPro ERP system. Identifying stakeholders helps ensure that the system requirements reflect the needs and expectations of all relevant parties involved in the project.

The following table lists the primary and secondary stakeholders associated with the ConstructPro ERP system.

ID	Stakeholder	Type	Role	Interest in the System
ST-01	Management (Ishara Homes)	Primary	Organizational decision makers	Monitor revenue, project performance, and business KPIs through analytics dashboards.
ST-02	Sales Team	Primary	Lead and quotation management	Capture client leads and generates project quotations efficiently.
ST-03	Project Managers	Primary	Project supervision	Track project milestones and monitor construction progress.
ST-04	Finance Officers	Primary	Financial operations	Record payments, manage invoices, and track outstanding balances.
ST-05	System Administrator	Primary	System management	Manage user accounts, roles, and system configuration.
ST-06	Development Team	Secondary	System development	Design, implement, and maintain the ConstructPro ERP system.

ST-07	Academic Supervisor	Secondary	Project oversight	Ensure the project meets academic standards and requirements.
ST-08	Clients (customers of Ishara Homes)	Secondary	Client portal users	View the progress, quotations, invoices, payments and documents of their own projects.

Table 2: Stakeholders

Chapter 8 Functional Requirements

Functional requirements define the specific operations and functionalities that the ConstructPro ERP system must perform to support the business processes of Ishara Homes Pvt. Ltd. These requirements describe how the system interacts with users and how it manages key activities such as lead management, quotation generation, project tracking, financial transactions, document management, and analytics reporting. Each requirement is assigned a unique identifier, priority level, and source stakeholder to ensure traceability, clarity, and proper validation throughout the software development lifecycle.

ID	Description	Priority	Source	Status
FR-001	Admin shall create and manage users	High	Management	Implemented
FR-002	System shall manage sales leads	High	Sales Team	Implemented
FR-003	System shall generate quotations	High	Sales Team	Implemented
FR-004	Quotation shall convert into projects	High	Management	Implemented
FR-005	Project milestones shall be tracked	High	Project Manager	Implemented
FR-006	The system shall track payments	High	Finance	Implemented

FR-007	The system shall provide KPI dashboard	Medium	Management	Implemented
FR-008	System shall store project documents	Medium	Project Manager	Implemented
FR-009	System shall integrate Google Calendar	Medium	Management	Implemented
FR-010	System shall provide AI risk prediction	Medium	Management	Implemented
FR-011	System shall provide a read-only client portal	Medium	Management	Implemented
FR-012	Project tasks shall be assigned and tracked	Medium	Project Manager	Implemented
FR-013	Project expenses shall be recorded	Medium	Finance	Implemented
FR-014	Users shall be able to sign in with Google	Low	Management	Implemented

Table 3: Functional Requirements

Chapter 9 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints of the ConstructPro ERP system. These requirements specify how the system should perform under various conditions and ensure that it remains reliable, secure, scalable, and maintainable. The following requirements are categorized according to key quality attributes such as performance, reliability, usability, security, scalability, maintainability, and portability.

ID	Category	Requirement	Acceptance Criteria
NFR-01	Performance	The system shall process user requests efficiently.	95% of user requests shall be responded to within 2 seconds under normal system load.
NFR-02	Performance	The system shall retrieve dashboard data efficiently.	Dashboard data retrieval operations shall be completed within 3 seconds.
NFR-03	Reliability	The system shall maintain consistent system availability.	System uptime shall be at least 99% during operational hours.
NFR-04	Reliability	The system shall support error monitoring and logging.	Automatic error logging shall record system failures for troubleshooting.
NFR-05	Usability	The system interface shall be accessible across devices.	The user interface shall be responsive on desktop, tablet, and mobile browsers.
NFR-06	Usability	Core system functions shall be easily accessible.	Lead creation and quotation generation shall be accessible within three clicks from the dashboard.

NFR-07	Security	The system shall implement secure user authentication.	Authentication shall use JSON Web Token (JWT) based authentication.
NFR-08	Security	The system shall restrict access based on user roles.	Access to system modules shall be controlled using Role-Based Access Control (RBAC).
NFR-09	Scalability	The system shall support increasing system load.	The architecture shall support horizontal scaling through cloud deployment.
NFR-10	Scalability	The system shall support multiple concurrent users.	The system shall handle at least 200 concurrent users without performance degradation.
NFR-11	Maintainability	The system shall support easy maintenance and updates.	Backend services shall follow modular architecture and coding standards.
NFR-12	Maintainability	The system shall allow safe system updates.	Updates and bug fixes shall be deployable without affecting core operations.
NFR-13	Portability	The system shall support deployment across cloud platforms.	The application shall be deployable on cloud platforms such as Vercel or AWS.
NFR-14	Portability	The system shall support modern web browsers.	The system shall function correctly on Chrome, Edge, and Firefox browsers.

Table 4: Non-Functional Requirements

Chapter 10 Business Rules

Business rules define the policies and operational constraints that govern how the ConstructPro ERP system functions within the business environment of Ishara Homes.

10.1 User Access Control

The system shall implement role-based access control to ensure that users only access information relevant to their responsibilities. Administrators will have full system privileges, while sales staff, project managers, and finance officers will have limited access based on their assigned roles.

10.2 Quotation Approval Policy

Quotations generated for clients must be approved by management before they can be converted into active projects. This ensures that project commitments are reviewed and verified before work begins.

10.3 Payment Recording Rule

All client payments must be recorded by the finance department before an invoice can be marked as completed in the system. This rule ensures financial records remain accurate and consistent.

10.4 Project Milestone Management

Each project must include defined milestones that represent important phases of the project lifecycle. Project managers are responsible for updating milestone progress to maintain accurate project tracking.

10.5 Client Portal Access

A client portal user can only view the projects, quotations, milestones, invoices, payments and documents that belong to their own customer account, and cannot create, change or delete any record.

10.6 Quotations per Project

A project may hold several quotations, for example for extensions and variations of the same build. Each approved quotation can be converted to a project, or linked to one, only once.

Chapter 11 Constraints

Constraints represent the limitations and restrictions that influence the design and implementation of the system.

11.1 Technology Constraints

The ConstructPro ERP system will be developed using specific technologies including Next.js for the frontend, Node.js with Nest.js for the backend, and PostgreSQL as the primary database. These technologies were selected to support scalability, flexibility, and modern web application development.

11.2 Integration Constraints

The system will integrate with external services such as the Google Calendar API for meeting scheduling, Google OAuth for sign-in, Cloudinary for document storage and OpenRouter for the AI module. Integration must comply with the technical specifications and authentication requirements provided by the external service.

11.3 Security Constraints

The system must implement secure authentication mechanisms such as JSON Web Tokens (JWT) to ensure that only authorized users can access system features and sensitive data.

11.4 Deployment Constraints

The system is expected to be deployed in a cloud-based environment to support remote access and scalability. Deployment environments may include Vercel cloud infrastructure services.

Chapter 12 Assumptions

Assumptions describe conditions that are expected to be true during system development and operation.

12.1 User Training

It is assumed that employees of Ishara Homes will receive basic training on how to use the ConstructPro ERP system before it is deployed for operational use.

12.2 Internet Connectivity

The system is designed as a web-based application, therefore it assumes that users will have reliable internet connectivity to access system features.

12.3 Availability of Historical Data

For the AI-based predictive analysis module to function effectively, it is assumed that sufficient historical project and financial data will be available within the ERP system.

12.4 Organizational Support

Successful implementation of the system assumes that management will support the adoption of the ERP platform and encourage staff members to use the system consistently for business operations.

Chapter 13 Use Case Diagrams

13.1 User & Role Management Use Case Diagram

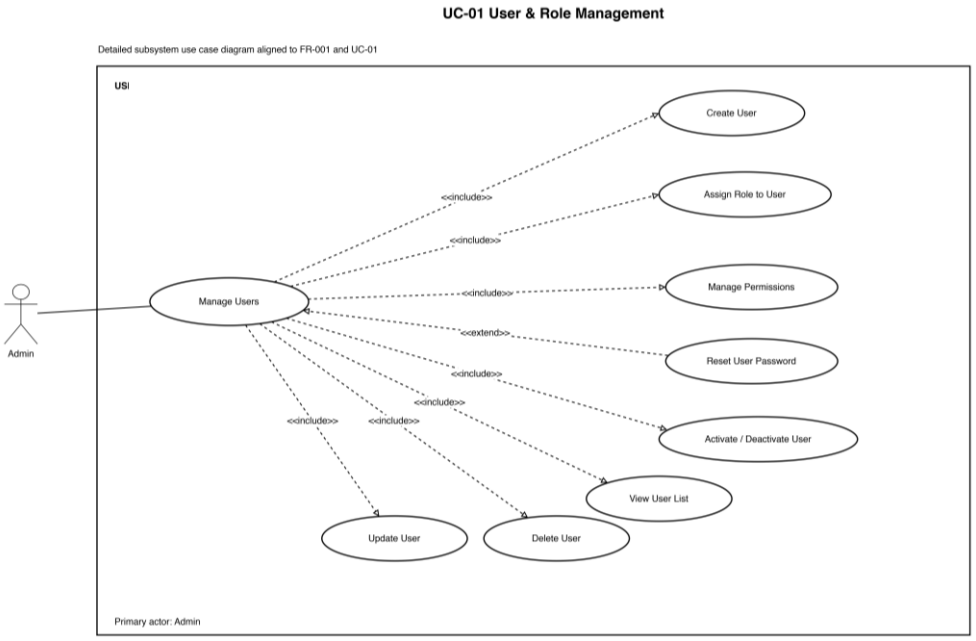


Figure 1: User & Role Management Use Case Diagram

13.2 Lead & Quotation Management Use Case Diagram

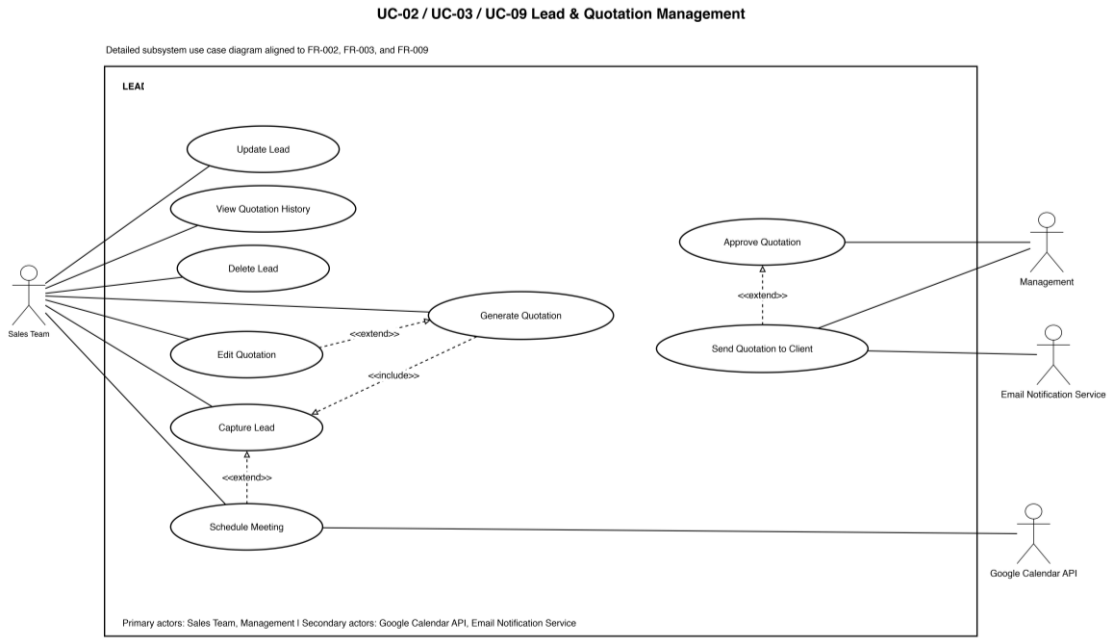


Figure 2: Lead & Quotation Management Use Case Diagram

13.3 Project Management Use Case Diagram

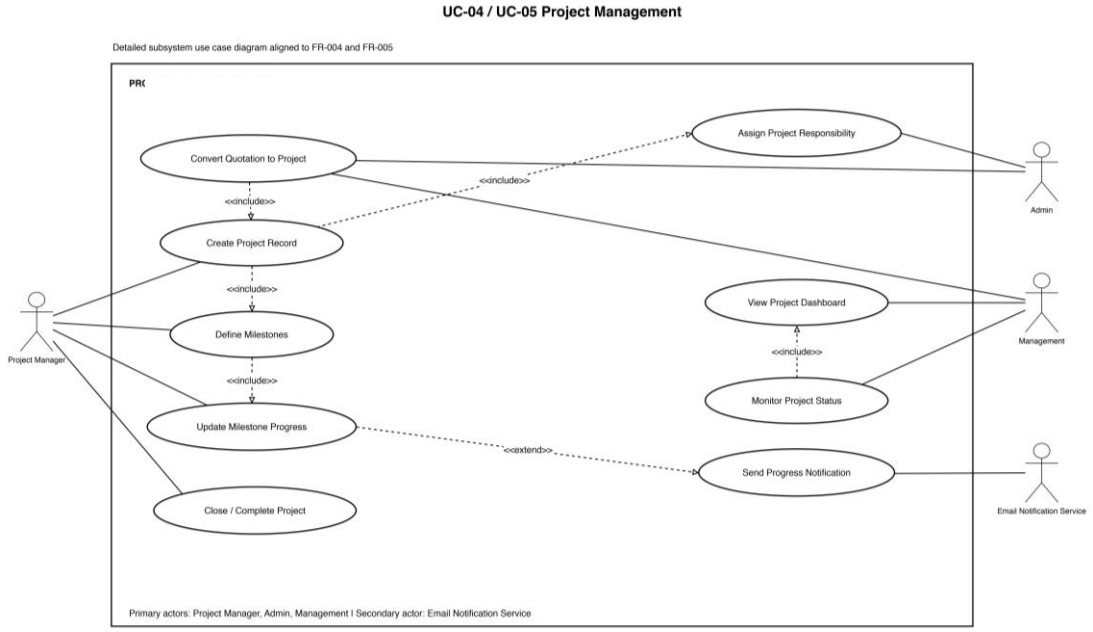


Figure 3: Project Management Use Case Diagram

13.4 Finance Management Use Case Diagram

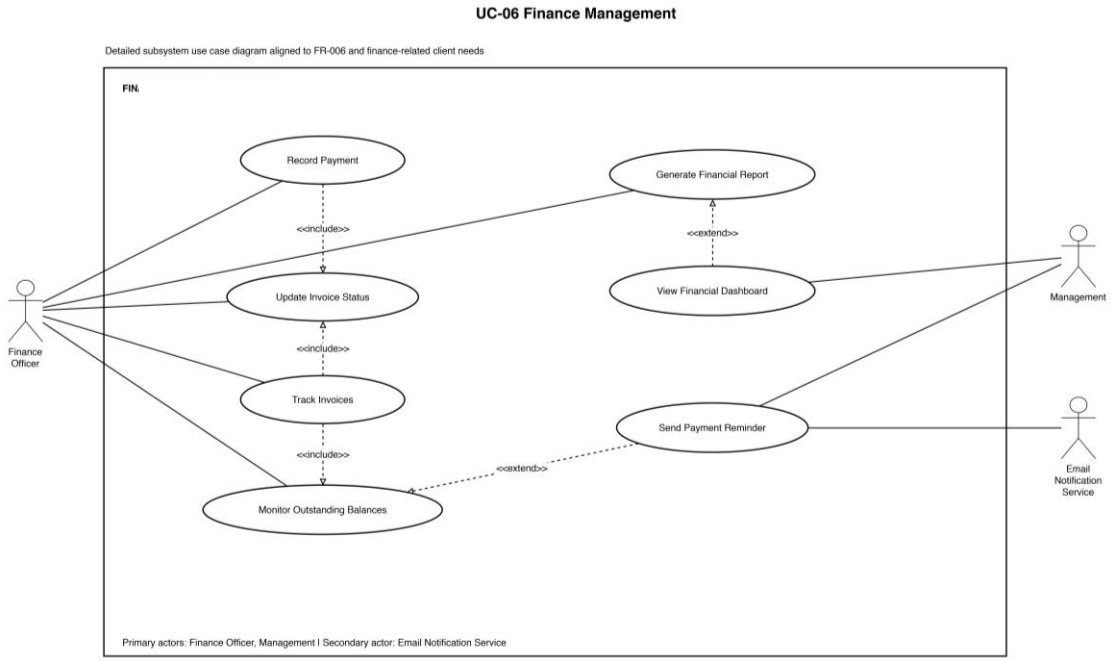


Figure 4: Finance Management Use Case Diagram

13.5 Analytics & AI Prediction Use Case Diagram

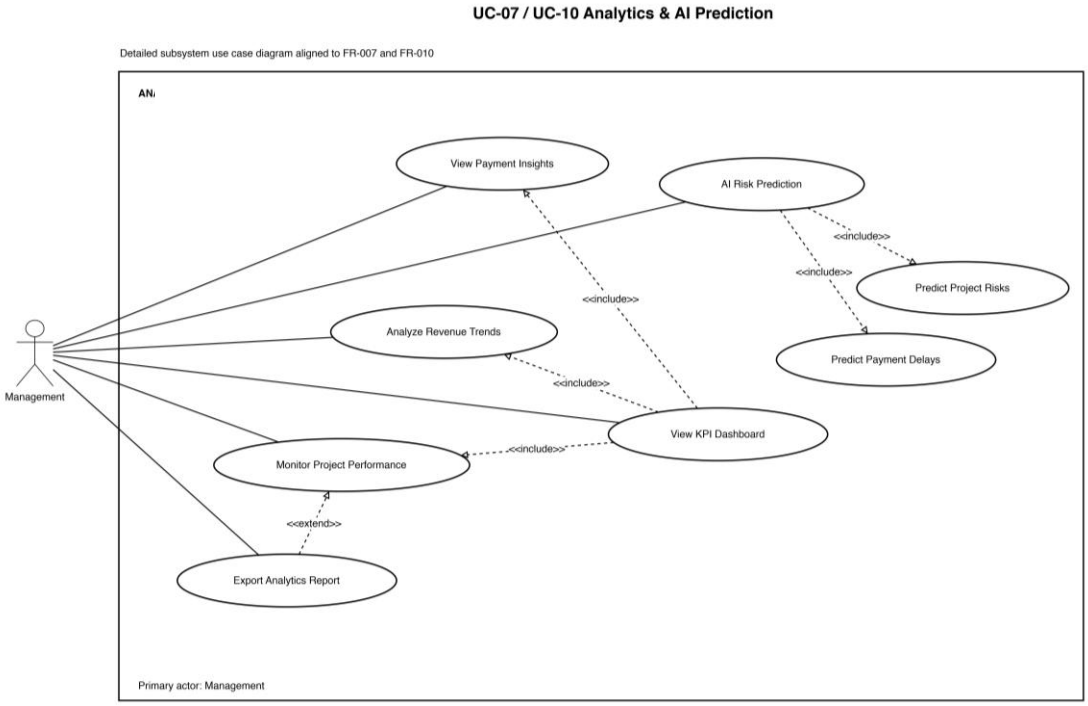


Figure 5: Analytics & AI Prediction Use Case Diagram

13.6 Document Management Use Case Diagram

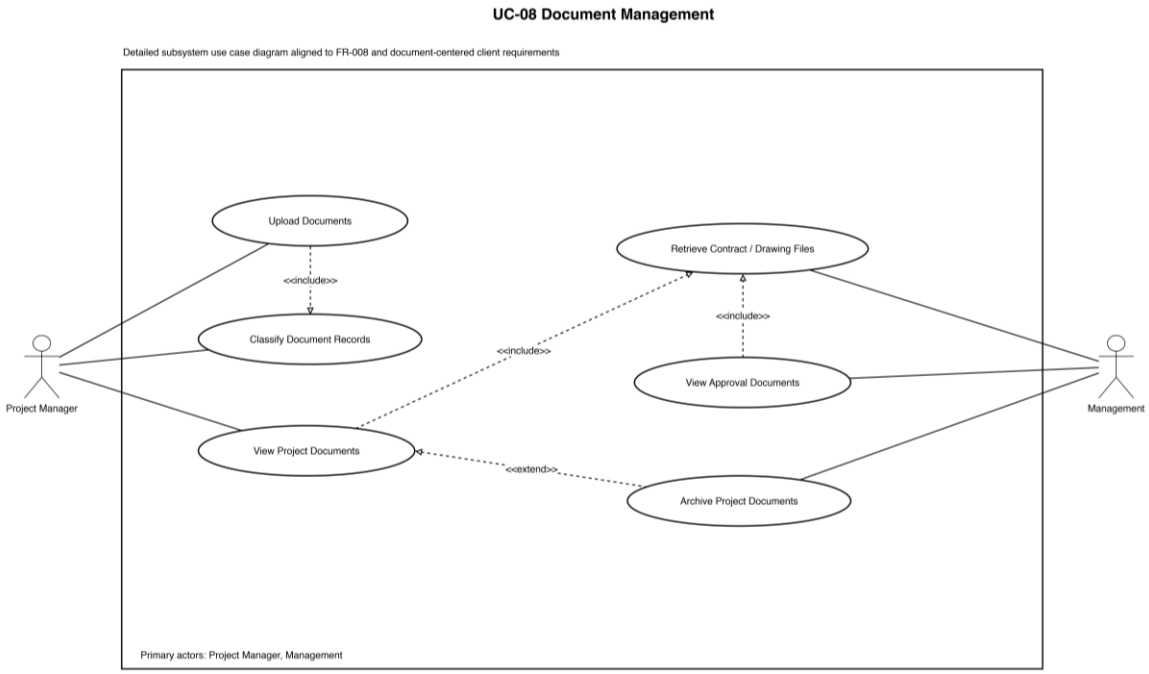


Figure 6: Document Management Use Case Diagram

Chapter 14 Use Case Descriptions

14.1 UC 01 — Manage Users

Field	Description
Use Case ID	UC-01
Use Case Name	Manage Users
Primary Actor	Admin
Description	Allows the administrator to create, update, delete, and manage system users and assign roles.
Preconditions	Admin must be authenticated and authorized.
Postconditions	User accounts and roles are successfully stored in the system database.
Main Flow	1. Admin logs into the system. 2. Admin opens the User Management module. 3. Admin creates or edits a user. 4. Admin assigns a role. 5. System saves the user information.
Alternative Flow	If user email already exists, the system displays an error message.

Table 5: UC01 - Manage Users

14.2 UC 02 — Capture Sales Lead

Field	Description
Use Case ID	UC-02
Use Case Name	Capture Lead
Primary Actor	Sales Team
Description	Sales staff record potential customer information as a lead in the system.
Preconditions	Sales user must be logged in.
Postconditions	Lead is stored in the database.
Main Flow	1. Sales opens Lead Management. 2. Sales enters client information. 3. Sales submits the lead form. 4. System stores the lead data.
Alternative Flow	If required information is missing, the system prompts the user to complete fields.

Table 6: UC02 - Capture Sales Lead

14.3 UC-03 — Generate Quotation

Field	Description
Use Case ID	UC-03
Use Case Name	Generate Quotation
Primary Actor	Sales Team

Description	Sales team generates a quotation based on a lead's requirements.
Preconditions	Lead must exist in the system.
Postconditions	Quotation is saved and ready for client review.
Main Flow	1. Sales selects a lead. 2. Sales enters quotation details. 3. Sales submits quotation. 4. System saves the quotation.
Alternative Flow	If quotation information is incomplete, the system prevents submission.

Table 7: UC03 - Generate Quotation

14.4 UC-04 — Convert Quotation to Project

Field	**Description**
Use Case ID	UC-04
Use Case Name	Convert Quotation to Project
Primary Actor	Management / Admin
Description	Converts an approved quotation into an active construction project.
Preconditions	Quotation must be approved by the client.
Postconditions	Project is created in the project management module.
Main Flow	1. Manager opens approved quotation. 2. Manager selects "Convert to Project". 3. System creates project record.

Table 8: UC04 - Convert Quotation to Project

14.5 UC-05 — Manage Project Milestones

Field	Description
Use Case ID	UC-05
Use Case Name	Manage Project Milestones
Primary Actor	Project Manager
Description	Project managers create and track project milestones and update progress.
Preconditions	Project must exist.
Postconditions	Milestones are recorded and progress is updated.
Main Flow	1. Project manager selects project. 2. Defines milestones. 3. Updates milestone status. 4. System updates project progress dashboard.

Table 9: UC05 - Manage Project Milestones

14.6 UC-06 — Record Client Payment

Field	Description
Use Case ID	UC-06
Use Case Name	Record Payment
Primary Actor	Finance Officer
Description	Finance team records client payments and updates invoice status.

Preconditions	Invoice must exist.
Postconditions	Payment is recorded and invoice status updated.
Main Flow	1. Finance opens invoice. 2. Enters payment amount. 3. Saves transaction. 4. System updates payment status.

Table 10: UC06 - Record Client Payment

14.7 UC-07 — View KPI Dashboard

Field	**Description**
Use Case ID	UC-07
Use Case Name	View KPI Dashboard
Primary Actor	Management
Description	Management views performance indicators such as revenue, payments, and project completion rates.
Preconditions	Manager must be logged in.
Postconditions	KPI data is displayed on the dashboard.
Main Flow	1. Manager logs in. 2. Opens Dashboard. 3. System loads KPI statistics. 4. Manager analyzes results.

Table 11: UC07 - View KPI Dashboard

14.8 UC-08 — Upload Project Documents

Field	Description
Use Case ID	UC-08
Use Case Name	Upload Documents
Primary Actor	Project Manager
Description	Project managers upload project-related files such as contracts, drawings, and approvals.
Preconditions	Project must exist.
Postconditions	Document is stored securely in the system.
Main Flow	1. Manager opens project. 2. Selects Upload Document. 3. Uploads file. 4. System stores the document.

Table 12: UC08 - Upload Project Documents

14.9 UC-09 — Schedule Meeting

Field	Description
Use Case ID	UC-09
Use Case Name	Schedule Meeting
Primary Actor	Sales Team

Description	Sales team schedules client meetings using Google Calendar integration.
Preconditions	Sales user logged in.
Postconditions	Meeting event is created in Google Calendar.
Main Flow	1. Sales selects meeting option. 2. Enters date and time. 3. System sends event to Google Calendar.

Table 13: UC09 - Schedule Meeting

14.10 UC-10 — AI Risk Prediction

Field	**Description**
Use Case ID	UC-10
Use Case Name	AI Risk Prediction
Primary Actor	Management
Description	AI system analyzes ERP data using RAG architecture to predict project risks and payment delays.
Preconditions	ERP data must exist.
Postconditions	Prediction insights are displayed on analytics dashboard.
Main Flow	1. Manager opens analytics module. 2. System retrieves ERP data. 3. AI model analyzes data. 4. System displays risk predictions.

Table 14: UC10 - AI Risk Prediction

Chapter 15 Activity Diagrams

15.1 Lead, Quotation & Project Workflow

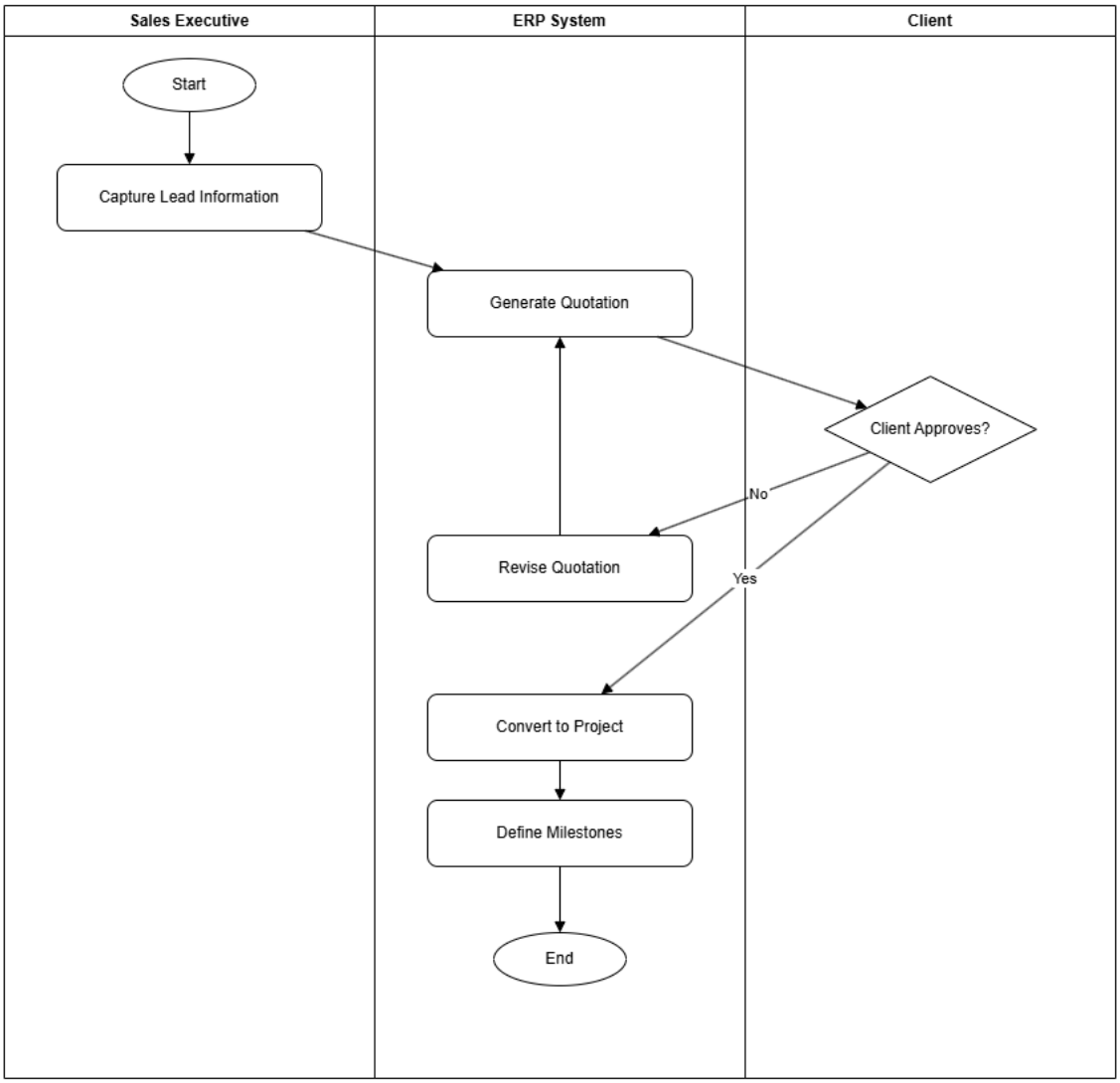


Figure 7: Lead, Quotation & Project Workflow

15.2 Project Management Workflow

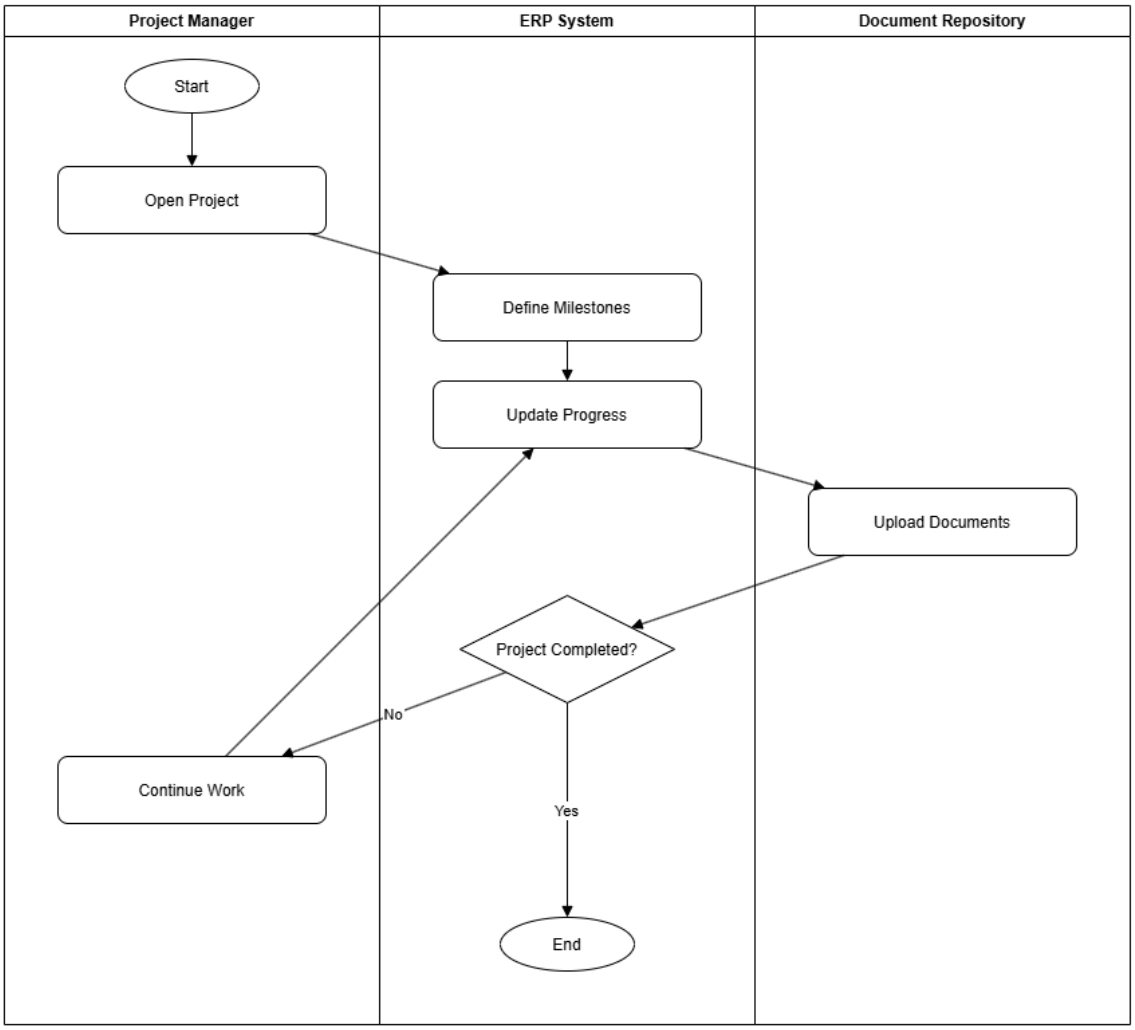


Figure 8: Project Management Workflow

15.3 Payment Management Activity

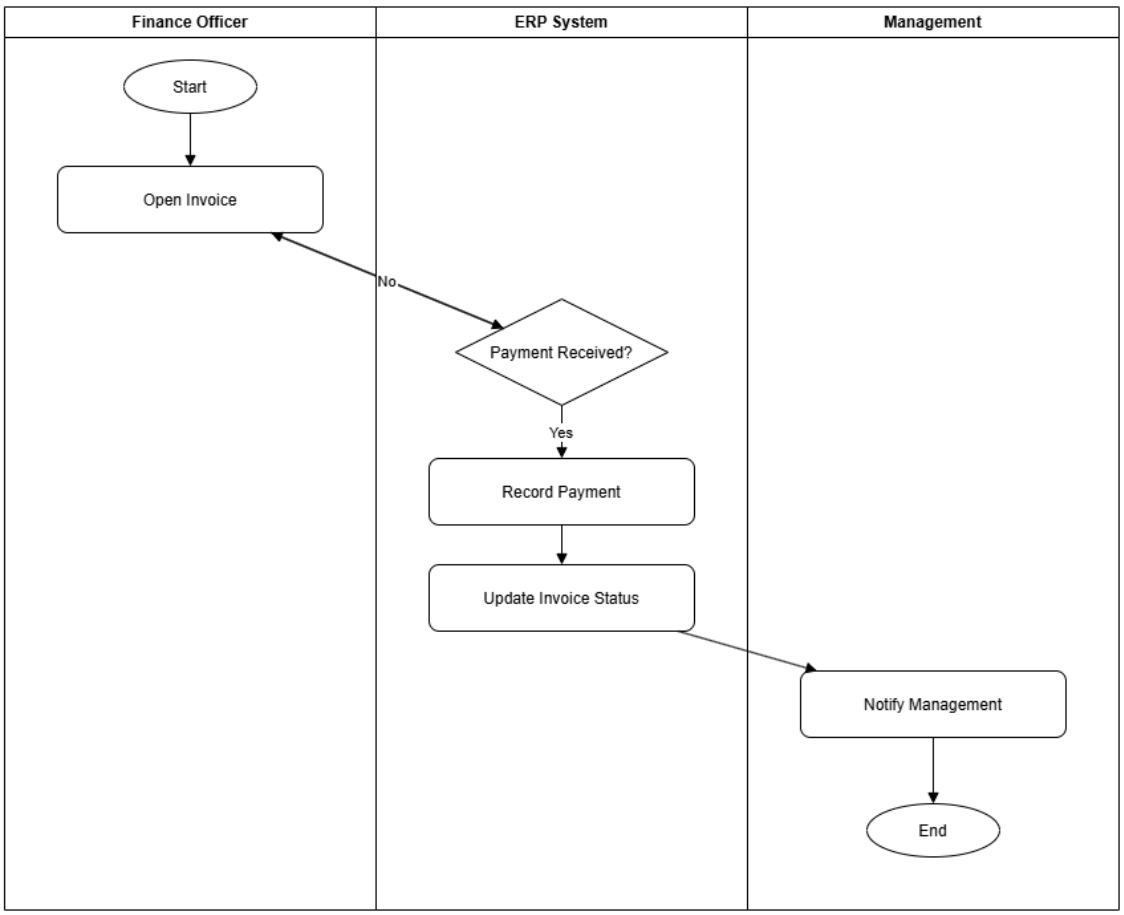


Figure 9: Payment Management Activity

15.4 KPI Dashboard Workflow

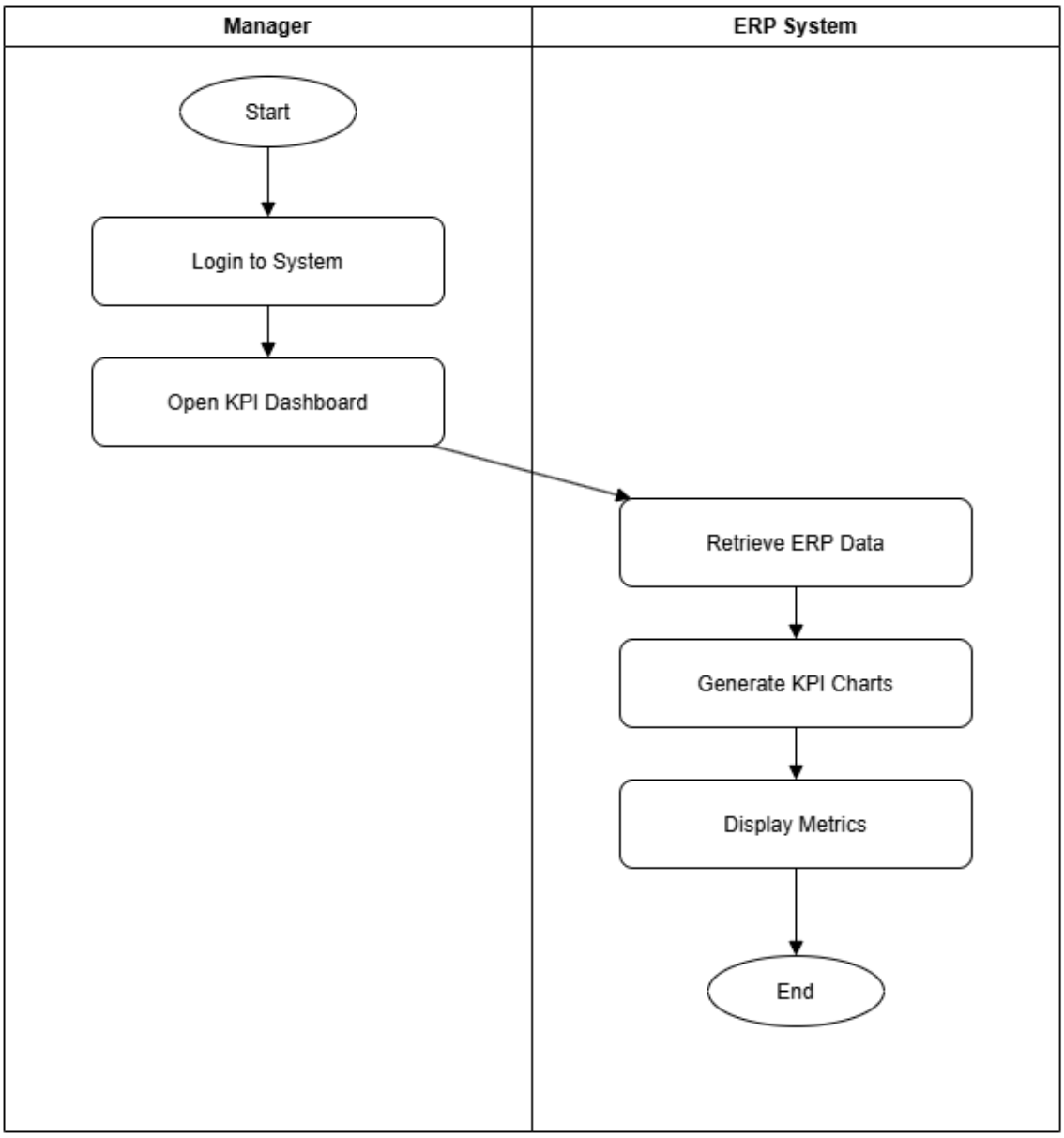


Figure 10: KPI Dashboard Workflow

15.5 AI Risk Prediction Workflow

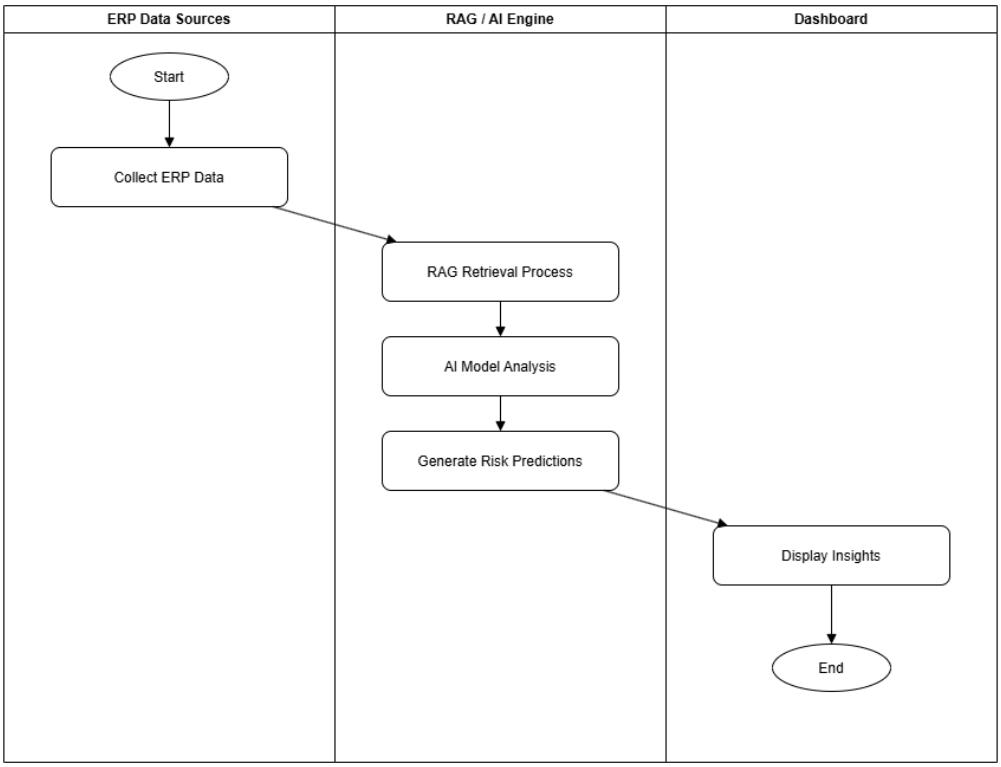


Figure 11: AI Risk Prediction Workflow

Chapter 16 External Interface Requirements

External interface requirements describe how the ConstructPro ERP system interacts with users, external systems, and communication networks.

16.1 User Interface Requirements

The system shall provide a user-friendly web interface that allows users to easily navigate and perform system operations. The interface will include modules for lead management, quotation generation, project tracking, financial monitoring, and analytics dashboards.

The user interface shall follow responsive design principles to ensure compatibility across desktop, tablet, and mobile devices. The frontend will be implemented using the Next.js framework, which supports modern web application development and optimized performance [3].

16.2 Hardware Interface Requirements

ConstructPro ERP is a web-based system and does not require specialized hardware. The system can be accessed using standard computing devices such as desktop computers, laptops, tablets, and smartphones. Users must have a device capable of running a modern web browser and maintaining an internet connection.

16.3 Software Interface Requirements

The system interacts with several external software services and technologies:

- Google Calendar API – used to schedule meetings and site visits directly from the ERP system [7].
- Google OAuth 2.0 – used for Google sign-in and for each user's Calendar connection [7].
- Cloudinary – used to store project documents and quotation PDFs.
- PostgreSQL database hosted on Neon – used to store system data such as users, projects, and financial records [6].

- Node.js and Nest.js – used to implement backend services and RESTful APIs [4], [5].

- OpenRouter API – used by the AI service to call a large language model for the RAG-based predictive analysis, with a rule-based fallback when the provider is unavailable [2].

16.4 Communication Interface Requirements

The system will communicate through standard web protocols. Data exchange between the frontend and backend services will occur using RESTful APIs over the HTTPS protocol. Data will be transmitted using JSON format to ensure efficient communication between system components.

Chapter 17 Data Requirements

The ConstructPro ERP system manages several types of data required for project management and operational monitoring.

17.1 Data Entities

The primary data entities managed by the system include:

- Roles
- User Statuses
- User
- Client
- Leads
- Quotations
- Projects
- Document Categories
- Documents
- Analytics Report
- Invoices
- Project Expenses
- Project Milestones
- Quotation Items
- Project Payments
- Project Tasks

These entities represent the core business information necessary to manage construction projects effectively.

17.2 Data Storage

All system data will be stored using the PostgreSQL database management system, which provides flexible document-based storage suitable for both structured and semi-structured project data [6].

17.3 Data Integrity

The system shall ensure data consistency and accuracy by implementing validation rules at both the application and database levels. Input validation will be performed before storing data, and database indexing will be applied to frequently accessed collections to improve query performance.

Chapter 18 Alternative Solutions Considered

Several alternative approaches were considered before selecting the ConstructPro ERP solution.

18.1 Manual Management Systems

The existing approach used by Ishara Homes involves managing project data through spreadsheets and manual documentation. Although this method requires minimal technical infrastructure, it is prone to data inconsistencies, duplication, and difficulty in tracking project progress.

18.2 Commercial ERP Systems

Commercial ERP platforms were also considered as potential solutions. While these systems provide a wide range of features, they are often expensive and may include functionalities that are not specifically tailored to the operational needs of Ishara Homes.

18.3 Custom ERP Solution

A custom-built ERP system was selected as the most suitable solution. Developing a dedicated system allows the organization to tailor the system to its business processes and integrate advanced features such as AI-based predictive analytics using a large language model through OpenRouter [2].

Chapter 19 Feasibility Study

The feasibility of the ConstructPro ERP system was evaluated based on technical, economic, operational, and schedule considerations.

19.1 Technical Feasibility

The system will be implemented using widely adopted web technologies, including Next.js, Node.js, Nest.js, and PostgreSQL. These technologies provide reliable frameworks for building scalable and maintainable web applications [3], [4], [5], [6].

19.2 Economic Feasibility

The project is economically feasible because it uses open-source technologies and cloud-based deployment environments, which reduce infrastructure and licensing costs. This approach allows the organization to implement the system without significant financial investment.

19.3 Operational Feasibility

The system is expected to improve operational efficiency by centralizing project management processes and improving information sharing between departments. Employees can adapt to the system with basic training and documentation.

19.4 Schedule Feasibility

The development of ConstructPro ERP is feasible within the timeframe allocated for the software engineering project. The project will be completed through planned development phases including requirement analysis, system design, implementation, and testing.

Chapter 20 Risk Analysis

Risk analysis identifies potential issues that may affect the successful development and operation of the ConstructPro ERP system. These risks are evaluated based on their likelihood and potential impact on the project. Appropriate mitigation strategies are proposed to reduce the probability and severity of these risks.

The risks identified in this project are categorized into technical, schedule, resource, requirement, and external risks.

20.1 Risk Register

Risk ID	Description	Likelihood	Impact	Risk Level	Mitigation Strategy
R-01	Misunderstanding of system requirements during development	Medium	High	High	Conduct regular meetings with stakeholders to validate requirements and update documentation.
R-02	Technical difficulties in integrating AI prediction module	Medium	Medium	Medium	Perform early testing of the RAG module and follow modular architecture for integration.
R-03	Data loss due to system failure or database issues	Low	High	Medium	Implement regular database backups and data recovery mechanisms.
R-04	Delays in project development due to limited development resources	Medium	Medium	Medium	Use clear project planning and divide tasks among team members effectively.

R-05	Security vulnerabilities or unauthorized access to sensitive data	Low	High	Medium	Implement secure authentication methods such as JWT and role-based access control.
R-06	Dependency on third-party services such as Google Calendar API	Low	Medium	Low	Monitor API updates and ensure fallback mechanisms if service is temporarily unavailable.

Table 15: Risk Register

20.2 Risk Categories

20.2.1 Technical Risks

Technical risks arise from challenges related to system development, integration, or technology limitations. For example, implementing the AI-based prediction system may require careful integration with the ERP database.

20.2.2 Schedule Risks

Schedule risks occur when development tasks take longer than expected. Delays in system implementation could affect the project timeline.

20.2.3 Resource Risks

Resource risks relate to the availability of team members, tools, or infrastructure required to complete the project.

20.2.4 Requirement Risks

Requirement risks occur when project requirements are unclear, incomplete, or frequently changing.

20.2.5 External Risks

External risks include dependencies on third-party services, internet connectivity, or organizational factors that may affect system operation.

Chapter 21 References

[1] M. Fowler, *Patterns of Enterprise Application Architecture*. Boston, MA, USA: Addison-Wesley, 2002.

[2] OpenRouter, "OpenRouter API Documentation." [Online]. Available: https://openrouter.ai/docs. Accessed: Oct. 23, 2026.

[3] Vercel, "Next.js Documentation." [Online]. Available: https://nextjs.org/docs. Accessed: Mar. 18, 2026.

[4] OpenJS Foundation, "Node.js Documentation." [Online]. Available: https://nodejs.org/en/docs. Accessed: Mar. 18, 2026.

[5] NestJS, "NestJS – A progressive Node.js framework." [Online]. Available: https://nestjs.com/. Accessed: Mar. 18, 2026.

[6] Neon, "Serverless Postgres: Neon database platform." [Online]. Available: https://neon.tech/. Accessed: Mar. 18, 2026.

[7] Google Developers, "Google Calendar API Documentation." [Online]. Available: https://developers.google.com/calendar. Accessed: Mar. 18, 2026.

Client Requirement Sign-off

The undersigned confirm that the Software Requirements Specification (SRS) for the ConstructPro ERP system has been reviewed and approved. This approval indicates agreement with the project scope, system requirements, and functional features described in this document for the development of the ConstructPro ERP platform for Ishara Homes Pvt. Ltd.

By signing this document, the stakeholders acknowledge that the requirements presented in this specification accurately represent the expected functionality and behavior of the system. The approved requirements will serve as the baseline for the subsequent phases of the software development lifecycle, including system design, implementation, testing, and deployment.

Any modifications to these requirements after approval must be submitted as a formal Change Request and must receive approval from both the client representative (Ishara Homes Pvt. Ltd.) and the project supervisor before implementation.

Client Representative Sign-off

Name: _____

Title: _____

Organization: _____

Date: _____

Signature: _____

Project Supervisor Sign-off

Name: _____

Date: _____

Signature: _____